Ex no: 08
22/9/26

Nmap Live Host Discovery ARP, ICMP and TCP/UDP

Aim:

To discover live hosts on a network using Nmap host discovery scans and to understand which protocol operate at which layer of TCP/IP model, and when each one applies.

ARP Host Discovery:

```
$ sudo nmap -PR -sn 10.200.6.0/24
```

O/P:-

```
Starting nmap 7.8 (https://nmap.org)
Nmap scan report for 10.200.6.1
Host is up
Nmap done: 256 IP addresses (1 host up)
scanned in 10.51 seconds.
```

ICMP Host discovery:

7.11 ICMP Echo (-PE)

```
$ sudo nmap -PE -sn 10.200.6.0/24
Nmap scan report for 10.200.6.1
Host is up (0.069s latency)
Nmap scan report for 10.200.6.250
Host is up (0.069s latency)
Nmap done: 256 IP addresses
scanned in 8.44 seconds
```

ICMP Timestamp (-PP)

$nmap -PP -sn 10.200-6.0124
Nmap scan report for 10.200.6.50
Host is up (0.069s latency)
Nmap scan report for 10.200.6.250
Host is up (0.069 latency
Nmap done: 256 IP addresses (2 hosts up)
scanned in 8:44 seconds.

ICMP Address Mask (-PM)

$ nmap -PM -sn 10.200.6.1024
Nmap done: 256 IP addresses (0 hosts up)
scanned in 52.14 seconds

TCP & UDP ping Discovery

i) TCP SVN Ping

$nmap -PS -SN 10.200.6.0124
Nmap scan report for 10.200.6.50
Host is up (0.70 s latency)
Nmap scan report for 10.200.6.250
Host is up (0.068s latency)
Nmap done: 256 IP addresses.

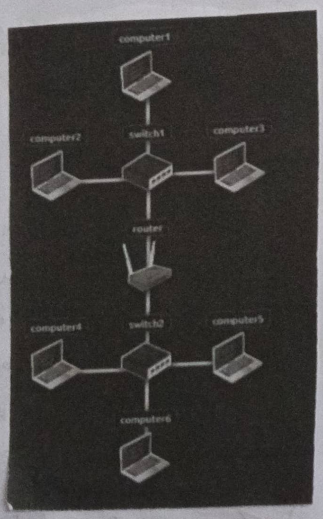
ii) TCP ACK Ping (-PA)

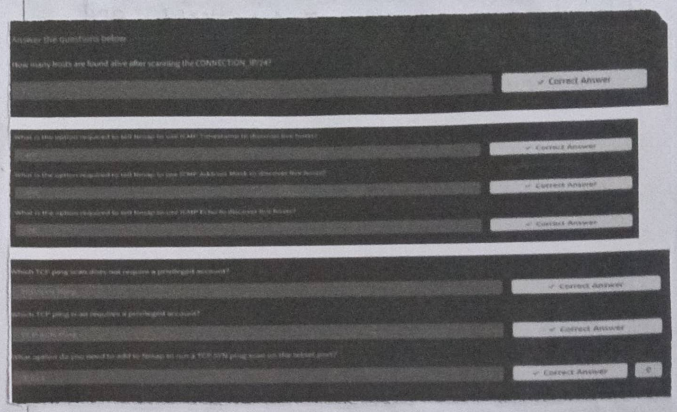
$ sudo nmap -PA -sn 10.200.6.0124
Nmap scan report for 10.200.6.50
Host is up (0.070s (latency)
Nmap scan report for 10.200.6.250
Host is up
Nmap done: 256 IP addresses

iii) UDP Ping (-PU)

$ sudo nmap -PU -sn 10.200.6.0/24

Nmap scan report for 10.200.6.52
Host is up (0.20s latency)
Nmap scan report for 10.200.6.250
Host is up (0.069s latency)
Nmap done: 256 IP addresses (2 hosts up) scanned in 35.97 seconds





Result:

Thus line host discovery for ARP, ICMP, TCP, UDP was studied and the o/p got verified successfully